

The Voice-AI Office: How Conversational Models Will Redesign Workplace Architecture

Executive summary

Office architecture has always been a physical expression of how work is organized: which roles need privacy, which teams must coordinate quickly, where decisions get made, and how knowledge spreads. In the hybrid era, that same logic has been strained by uneven attendance patterns and more “purposeful” days in the office, forcing companies to rebalance space toward collaboration, focus, and technology-rich meeting environments. Surveys of corporate real estate leaders show office attendance stabilizing into a “steady state,” with peak utilization concentrated mid-week and Fridays rarely the peak day—conditions that make flexibility and space variety core design requirements rather than “nice-to-haves.” ¹

Voice AI changes the equation because it makes *speech*—already the main acoustic disruptor in offices—the primary interface for capturing work, creating institutional memory, and controlling environments. Mainstream collaboration platforms now offer AI meeting recap and summarization built on transcription, increasingly with speaker attribution; enterprise voice assistants can start meetings and control devices with wake words; and multilingual speech recognition and translation are moving from captions to near-real-time speech-to-speech translation. ²

As a result, the “AI-ready” office is not just a better video-conferencing office. It is an office that deliberately **zones for “speakable work”** (places where talking—into meetings, agents, dictation, or translation—doesn’t disrupt others), **treats audio capture as a building utility** (like lighting and Wi-Fi), and **designs privacy into both space and systems** (clear “no-mic” zones, visible recording states, retention controls, and policy guardrails). This is especially critical because voice-based features can become “employee monitoring” in practice if not governed; regulators have issued guidance on monitoring workers, and the EU AI Act specifically prohibits certain workplace uses such as emotion inference, while requiring transparency and worker information obligations for some high-risk workplace deployments. ³

The near-term design shift is toward **more small, bookable, acoustically controlled rooms** (for 1–4 people) paired with **fewer but higher-quality, “hybrid-equity” meeting rooms** optimized for speaker recognition, captions, and multi-stream video. At the same time, the classic open-plan “communication advantage” continues to be constrained by well-documented dissatisfaction with noise and lack of speech privacy—precisely the conditions that reduce transcription quality and raise information leakage risks. ⁴

For leaders remodeling today, the winning approach is a **phased roadmap**: instrument a small set of “capture-ready” rooms first (to learn what policies and acoustics actually work), then expand with a standard “room kit of parts” (audio, privacy signals, and governance), while treating procurement and change management as seriously as construction. Use risk-management frameworks to align stakeholders (Facilities, IT, Security, HR, Legal) and to define what speech is captured, where it is stored, who can access it, and how long it persists. ⁵

Why organization design determines office architecture today

The starting point for workplace design is not furniture; it is **coordination costs**. Organizations divide work into roles (specialization) and then recombine it through workflows (integration). Classic organizational research distinguishes, for example, between **functional structures**, which group people by specialty, and **project-oriented structures**, which group people around shared tasks—each creating different communication needs and therefore different spatial needs. ⁶

In practice, this is why office layouts historically cluster roles into “neighborhoods” and use adjacency to reduce friction:

- **High-interdependence teams** (product engineering, R&D programs, incident response) benefit from frequent, low-latency interaction—hence co-location, shared team rooms, war rooms, and dense collaboration zones.
- **High-concentration work** (software development, analysis, writing, design production) needs protection from interruption—hence focus rooms, libraries/quiet zones, and private offices in some contexts.
- **Externally facing roles** (sales, recruiting, customer support) require “call-capable” environments with acoustic privacy, reliable AV, and quick access to meeting rooms or call pods.

This logic is reinforced by research on communication and proximity. Work on architecture and communication in technical organizations has long emphasized that physical space influences who interacts with whom and how information moves through an organization. ⁷

However, the modern office also reflects countervailing pressures: space efficiency, cost, and the hope that open-plan layouts increase interaction. Yet evidence consistently shows a trade-off: open-plan environments are commonly associated with dissatisfaction driven by **uncontrollable noise and loss of privacy**, even if they are intended to support communication. ⁸ A recent field study modeling acoustic dissatisfaction in open-plan offices found that “lack of privacy” was a stronger contributor than noise disturbance in predicting dissatisfaction—an important clue for any future office that depends on speech capture for AI.

⁹

The hybrid era has amplified the need to match space to work modes. Corporate real estate surveys show many organizations now treat office utilization as stable but uneven, planning around peaks and troughs rather than a uniform daily load. One major survey reports that more than 60% of occupiers said attendance had reached a steady state, and that peak utilization is concentrated mid-week (with Fridays rarely the peak). ¹⁰ This shifts design from “one desk per person” toward a portfolio of settings: unassigned seating, more small rooms, and more collaboration spaces—paired with policies that keep the office usable at peak occupancy without overbuilding for trough days. ¹¹

What voice AI can do now and what it changes

Voice AI is not one feature; it is a *stack* that turns spoken interaction into searchable, automatable work. The capabilities below are already shipping in major enterprise platforms, and emerging research is pushing them toward real-time and “agentic” behaviors.

Real-time transcription, speaker attribution, and meeting recap are rapidly becoming defaults. Microsoft Teams supports “Intelligent recap” (administrative guidance and requirements are documented), and Teams Rooms can use “Intelligent Speaker” voice recognition so transcripts identify who said what—explicitly positioning speaker identity as critical input for Copilot-style meeting summaries and insights. ¹² Zoom’s AI Companion can generate a meeting summary using speech-to-text as input, and Google Meet’s “take notes for me” attaches AI-generated notes to the calendar event, reducing the manual labor of note-taking and follow-up. ¹³ Academic work mirrors this trajectory: meeting summarization research highlights the unique challenges of dialogue and long context, while newer work explores real-time, agenda-aware meeting summarization with large language models. ¹⁴

Voice assistants and conversational agents are moving from phones into rooms. Cisco’s Webex Assistant for devices supports voice-driven commands for meetings and device controls, using a wake phrase (“OK Webex”) and offering actions like booking or extending a room—an early example of hands-free room operation. ¹⁵ The design implication is straightforward: rooms are becoming *interactive systems* that must be reliable for non-technical users, with microphones and speakers treated as core infrastructure.

Ambient listening is the most design-disruptive capability—not because buildings will record everything, but because always-available microphones and wake-word behavior change privacy expectations. A wake-word assistant implies a device is continuously listening for activation, even if it is not always recording; that distinction matters little to perceived privacy unless the environment provides clear signaling, controls, and policy transparency. ¹⁶

Multilingual speech recognition, translation, and speech-to-speech interpretation are shifting meetings from text captions to cross-language conversation. Microsoft documents multilingual speech recognition in Teams, including live translated captions/transcripts and translated recap after multilingual meetings. ¹⁷ Google Meet provides translated captions and, more recently, announced speech translation in Meet leveraging real-time speech-to-speech translation research. ¹⁸ Google Research describes an end-to-end speech-to-speech translation model enabling real-time translation in the original speaker’s voice with low delay—an indicator of where “meeting room language barriers” are headed. ¹⁹

Voice biometrics (for recognition, authentication, or speaker attribution) introduces a different class of risk. NIST runs speaker recognition evaluations and publishes work to measure and advance speaker recognition capabilities, underscoring that voice biometrics is an active field with evolving performance and operational best practices. ²⁰ At the same time, anti-spoofing research and challenges (e.g., ASVspoof) show the growing sophistication of speech deepfakes and spoofing attacks, and standards bodies publish frameworks for presentation attack detection. ²¹ This matters for offices because as soon as voice becomes an identity signal (for room access, meeting attribution, or device control), **physical space and security converge**: microphones, speakers, and room systems become part of the attack surface.

The AI-era office: new spatial archetypes

Voice AI pushes office design toward a new core question:

Where is it acceptable—and productive—for people to speak out loud to machines and to each other?

In the pre-AI office, speech was mostly a byproduct (calls, conversations, meetings). In the voice-AI office, speech becomes a *work method*: dictating drafts, interrogating a conversational agent, creating voice notes, and collaborating in multilingual settings. That increases the demand for **speech-capable spaces** in the same way laptops increased demand for power and Wi-Fi.

The four-zone framework

A practical way to translate voice AI into floorplate design is to plan for four “speech and capture” zones:

Quiet production zone (low speech)

A library-like area where speech is discouraged because it degrades focus and creates unwanted capture risk. This zone protects deep work and reduces the baseline noise floor that degrades adjacent transcription quality, aligning with evidence that acoustics and speech privacy drive satisfaction in open environments.

22

Voice pods and micro-rooms (controlled speech)

A high-density set of 1-person and 2-4 person rooms built for calls, dictation, and “talking to an agent.” Hybrid work surveys show increased emphasis on small rooms and focused-work spaces as firms adapt portfolios. ²³

23

Collaboration hubs (normal speech, low capture commitment)

Open or semi-open areas for impromptu collaboration where speech is expected but not necessarily recorded. These spaces need strong acoustic design to prevent speech spillover and to avoid turning the entire office into a “leaky meeting.” ISO 3382-3 specifically focuses on measuring open-plan office acoustic performance for scenarios where speech propagation and speech privacy are central concerns. 24

24

Capture-ready meeting rooms (high capture commitment)

Rooms intentionally designed for hybrid equity (camera framing, multi-stream views, high-quality microphones) and for AI features (transcription, attribution, summarization). Microsoft's meeting-room guidance for Teams and its Front Row layout illustrate how room geometry and furniture placement are now designed around remote/in-room parity rather than around a single "conference table." ²⁵ Zoom Rooms Smart Gallery similarly uses multiple streams from one room to make in-room participants more legible to remote attendees. ²⁶

26

Example floorplate concept

Below is a simplified “AI voice-ready” floorplate pattern (not a prescriptive plan, but a design aid). It assumes you want to keep the quietest work away from collaboration hubs and keep “capture-ready” rooms near strong IT/AV risers for reliable performance.

```
+-----+
| QUIET PRODUCTION ZONE (low speech) |
| - focus desks + silent nooks + heads-down rooms |
|                                     |
| [F] [F] [F]          acoustic buffer          [F] [F] [F] |
|                                     |
+-----+
```

-----+-----+-----		
VOICE	COLLAB HUB (normal	CAPTURE-READY ROOMS
PODS	speech / low capture)	(high capture commitment)
(1-2p)	lounges + boards	4p / 8p / 12p
[P][P]	+ quick huddle spots	+ “hybrid equity” setups
[P][P]		[M] [M] [M]
-----+-----+-----		
“TELEMETRY & SERVICES SPINE” (IT closets, AV racks, sensors)		
+-----+-----+-----+		

This concept reflects a key operational reality from hybrid utilization data: you need enough small rooms and flexible zones to absorb mid-week peaks without destroying focus conditions everywhere else. ¹⁰

Design options and trade-offs

Design choice	What it optimizes	What it risks	When it works best
“AI everywhere”: many spaces always mic-enabled for capture	Maximum convenience; frictionless transcription	Perceived surveillance; speech leakage; sensitive conversations captured unintentionally; harder compliance posture (worker monitoring concerns) ²⁷	Low-regulated environments; high trust culture; clear opt-in/opt-out and strong transparency
“Capture islands”: a limited set of clearly labeled capture-ready rooms	High governance clarity; easier user education and signage	Users may avoid AI features if rooms are scarce; booking contention at peak times	Most enterprises starting adoption; mixed sensitivity workloads; early rollouts
“Pods-first”: invest heavily in 1–2 person booths for calls/agents	Supports voice workflows without disrupting open areas	If overdone, can reduce social cohesion; may feel like call-centerization	Engineering-heavy or writing-heavy orgs; hybrid hot-desking; high call volume teams
“Hybrid boardrooms”: fewer rooms but very high quality (multi-stream video + premium audio)	Executive-grade meetings, external calls, multilingual gatherings	Underutilization when attendance is uneven; high capex per room	Client-facing orgs; global teams; high-stakes meetings needing translation and recap ²⁸

What changes in specific space types

Hybrid work and hot-desking

Hybrid attendance stability plus uneven peaks pushes offices toward unassigned seating and higher space variety. One major occupier survey explicitly frames utilization planning around peak/average/trough metrics and suggests offices feel “full” around ~85% utilization for unassigned workplaces—highlighting

why “just add more desks” is rarely the right answer. ¹⁰ Voice AI amplifies the need for **personal storage and identity continuity** even when desks are shared: lockers, quick device pairing, and consistent acoustic options (pods) become part of the employee experience rather than “amenities.”

Meeting rooms

Meeting rooms become *compute rooms*: voice quality determines recap quality, and recap quality determines downstream productivity. Microsoft’s guidance emphasizes layouts and specifications for Teams Rooms and hybrid meeting experiences, while voice recognition documentation ties speaker identity directly to Copilot-quality meeting outcomes. ²⁹ Zoom’s Smart Gallery highlights a similar “equity” objective (making in-room participants legible remotely). ³⁰ The design implication is that the “minimum viable meeting room” rises: if audio is poor, AI output will be poor, and users will abandon the features.

Focus rooms

Focus rooms become not only “quiet rooms” but also **privacy boundary rooms**—places where users can safely ask an agent questions involving sensitive topics (performance feedback drafts, legal questions, negotiations). This intersects directly with worker monitoring concerns and the need for proportional, transparent capture. ²⁷

Collaboration hubs

Collaboration hubs remain essential for social cohesion and creativity, but they must be acoustically engineered so speech doesn’t propagate across the floor. ISO 3382-3 exists precisely because open-plan offices require objective measures for how speech decays with distance and how speech privacy performs in a space. ³¹ Voice AI makes the downside of poor acoustics more expensive: it harms both human focus and machine transcription.

Infrastructure, acoustics, and privacy-by-design

A voice-AI workplace fails for three predictable reasons: (1) inadequate acoustics, (2) brittle AV/IT infrastructure, and (3) unclear governance that triggers distrust or regulatory risk. Designing for voice AI therefore requires treating **audio + policy** as a coupled system.

Acoustics becomes a “model performance” variable

Open-plan dissatisfaction research consistently points to noise and lack of sound privacy as key problems, and newer modeling work links “lack of privacy” strongly to acoustic dissatisfaction. ²² For voice AI, acoustics is not just comfort—it is **input quality**. Better speech-to-text, diarization, and translation rely on clear capture, which depends on room geometry, materials, and microphone placement.

A practical, standards-aligned approach is to:

- Use open-plan acoustic measurement concepts and targets anchored in standards like ISO 3382-3 (focused on open-plan office acoustic parameters and speech privacy behavior). ³²
- Concentrate “high-capture” capabilities in rooms designed as *acoustic systems*: controlled reverberation, isolation, and appropriate microphone arrays (often ceiling or table arrays in modern room kits). Microsoft’s Teams room planning guidance reflects that room type and layout drive technology specifications. ³³

AV/IT backbone and building integrations

Voice AI expands the “digital perimeter” of the office:

- **Network reliability** matters more because audio/video streams, live captions, and AI services are sensitive to jitter and packet loss; vendors explicitly publish bandwidth/network requirements for meeting hardware. ³⁴
- **Room systems become platform endpoints:** Teams Rooms, Zoom Rooms, and Meet room kits are no longer peripherals—they are managed devices requiring lifecycle management, patching, and security posture. ³⁵
- **Building controls enter the voice surface area** as assistants evolve from “meeting control” to “room control.” BACnet (ASHRAE Standard 135) is a widely used vendor-independent protocol for building automation, covering applications such as HVAC, lighting, access control, and fire detection—exactly the systems voice-driven controls tend to touch. ³⁶

In other words: once occupants can ask rooms to *do things*, the office needs an explicit **integration architecture** (platform endpoints, identity, network segmentation, logging, and fail-safes).

Security and privacy are spatial issues now

Voice AI introduces new categories of personal data and new governance expectations:

- If voice is used to uniquely identify a person (voiceprints for recognition), it can become **biometric data**, which is treated as a special category under GDPR when used for unique identification. ³⁷
- The EU AI Act prohibits certain practices in workplace contexts (notably, emotion inference in workplaces and education, with limited exceptions). ³⁸
- The EU AI Act also establishes requirements around informing workers before using certain high-risk AI systems at the workplace and lists “employment, workers’ management and access to self-employment” as a high-risk area in Annex III. ³⁹
- Worker monitoring guidance emphasizes lawful basis, transparency, and proportionality; the UK ICO provides guidance and tools for employers considering employee monitoring (which can include audio recording). ²⁷

Meanwhile, voice biometrics is an active security domain: NIST speaker recognition programs and evaluations highlight ongoing measurement of system performance, while presentation attack detection standards and anti-spoofing challenges reflect the reality of spoofing and deepfake threats. ⁴⁰

Design patterns that connect space and governance

Pattern	Spatial implementation	Tech implementation	Where it reduces risk
“Capture labeling”	Clear signage: <i>AI-enabled capture room vs no-capture room</i> ; visible “recording” indicators	Platform settings enforce transcription/recording defaults by room type	Lowers perceived surveillance; reduces accidental capture ⁴¹

Pattern	Spatial implementation	Tech implementation	Where it reduces risk
“Push-to-capture” default	In open areas, avoid ambient capture; concentrate capture in enclosed rooms	Meeting host/room policy requires explicit enablement for summaries/recordings	Aligns with proportionality expectations for monitoring ⁴²
“Privacy tiers”	Tier 0: no mics; Tier 1: mics for live audio only; Tier 2: transcription; Tier 3: attribution + recap	Data retention, access control, and audit trails increase with tier	Enables sensitive teams (R&D, HR) to use space without over-collection ⁴³
“Biometric opt-in”	Provide alternative rooms/workflows for guests/non-enrolled users	Voice profile enrollment is optional; speaker attribution works only when opted in	Reduces biometric friction and supports compliance expectations ⁴³
“Model-risk governance”	Rooms designed for policy compliance (e.g., HR/Legal rooms not AI-enabled by default)	Apply AI and privacy risk frameworks to define acceptable use	Creates shared language across Facilities/IT/HR/Legal ⁴⁴

Roadmap for remodels and the next wave of trends

A remodel that “future-proofs for voice AI” is less about betting on one vendor and more about building **reusable capabilities**: acoustics, room typologies, governance, and an upgradable backbone.

A phased roadmap that works under uncertainty

Phase one: Strategy and baselining (0–3 months)

Define the work patterns the office must support (by team type), then audit current pain points: room shortage, poor audio, lack of privacy, uneven utilization. Hybrid utilization guidance suggests measuring peak/average/trough and planning accordingly rather than relying on average occupancy alone. ¹⁰ At the same time, establish an AI governance baseline using widely adopted risk frameworks (NIST AI RMF for AI risk management; NIST Privacy Framework for privacy risk as enterprise risk). ⁴⁴

Phase two: Pilot “capture-ready” standards (3–9 months)

Select a small number of rooms across sizes (e.g., 2p, 4p, 8–10p) and implement a “room kit of parts”:

- high-quality microphones/speakers suitable for transcription and speaker recognition (where desired) ⁴⁵
- hybrid-equity layout features (Front Row-like layouts or equivalent) ⁴⁶
- explicit user signals (recording indicators, signage, room tier labels) ⁴¹
- policy defaults (what is allowed in that room, retention expectations, who can enable what)

Then run the pilot long enough to observe behavior under peak days, not just averages. ¹⁰

Phase three: Scale space types, not one-off rooms (9–24 months)

Expand using repeatable typologies: more micro-rooms, a balanced mix of collaboration hubs, and fewer but better large rooms. Align with known dissatisfaction drivers: prioritize speech privacy and noise management rather than assuming open-plan will “self-regulate.” 22

Phase four: Integrate building systems and advanced voice control (24+ months)

Once trust and governance are stable, add voice-driven room/environment control where appropriate (booking, starting meetings, device control), and integrate with building automation through standard protocols (e.g., BACnet/ASHRAE 135) using network segmentation, logging, and security review. 47

Decision flowchart for “what do we enable where?”

```
flowchart TD
    A[Start: Define room or zone] --> B{Is speech expected here?}
    B -->|No| C[Quiet / No-mic zone\n- No voice assistants\n- No transcription defaults]
    B -->|Yes| D{Is confidentiality high?}
    D -->|High| E[Private room tier\n- Strong isolation\n- Push-to-capture\n- No ambient listening]
    D -->|Medium| F{Is AI recap needed?}
    F -->|Yes| G[Capture-ready room\n- High-quality mics\n- Visible recording state\n- Retention + access controls]
    F -->|No| H[Collab space tier\n- Good acoustics\n- No default transcription]
    G --> I{Speaker attribution / voice biometrics allowed?}
    H --> I
    I -->|Opt-in only| J[Enable voice profiles with alternatives\n- Guests = generic speakers]
    I -->|Not allowed| K[Transcription without attribution]
```

This logic aligns with regulatory direction that emphasizes transparency and proportionality for workplace monitoring and with restrictions on certain workplace AI practices (e.g., emotion inference). 48

Recommended features by team type

Team type	Dominant work modes	Space mix to prioritize	Voice AI features to emphasize	Privacy/acoustics implications
Engineering	Focus work + standups + design reviews	Quiet production zone, many 1p pods, several 4–6p rooms, a few hybrid rooms	Transcription + summaries for standups/design reviews; voice-to-agent in pods; selective speaker attribution for action tracking 49	High need for speech containment so voice workflows don’t disrupt focus; open-plan speech dissatisfaction risk is well documented 22

Team type	Dominant work modes	Space mix to prioritize	Voice AI features to emphasize	Privacy/acoustics implications
Sales / BD	Calls, demos, customer meetings, follow-ups	Many call pods, small “client-ready” rooms, a few premium rooms	Meeting summaries/ action items; translation for global calls; voice assistant for fast room control ⁵⁰	Frequent speech means acoustic privacy is non-negotiable; clear capture labeling reduces client trust concerns ⁵¹
Customer support	High call volume, QA/ coaching, knowledge base loops	High-density pods, QA rooms, training rooms	Real-time transcription (internal), summarization into tickets KB; selective monitoring governance	Highest risk of “monitoring” perceptions; must align with monitoring guidance and lawful, transparent practices ⁴²
R&D / advanced research	Deep work + secure collaboration	Strong quiet zones, secure project rooms, limited capture-ready rooms	Summaries for internal reviews where allowed; avoid broad ambient capture; consider on-device/ edge translation in sensitive contexts	Often the highest confidentiality requirements; design “no-capture” rooms and strict tiering; biometrics governance is critical ⁵²

What will likely come next

From captions to real-time speech-to-speech translation as a default meeting layer

Google is already productizing speech translation in Meet and explicitly links it to real-time speech-to-speech translation research; Google Research describes low-latency S2ST that preserves the speaker’s voice. ⁵³ As speech translation becomes normal, meeting-room design will increasingly assume multilingual audio pipelines (better microphones, less reverberation, clearer turn-taking cues).

From “meeting summaries” to “meeting agents” that intervene in real time

Research is moving toward real-time, agenda-aware summarization, and industry is shipping more “live notes” and post-meeting automation. ⁵⁴ This will shift room UX toward visible, in-room AI states (what is being captured, summarized, or actioned).

From “smart rooms” to “smart building voice surfaces,” raising the stakes on cybersecurity

As voice control touches building systems, a vendor-independent automation backbone (BACnet/ASHRAE 135) becomes part of enterprise threat modeling, not just facilities integration. ⁵⁵

From authentication to anti-spoofing arms races

As voice becomes an identity factor, spoofing threats grow. Standards for presentation attack detection and

ongoing anti-spoofing research (ASVspoof) suggest that any voice-biometric-enabled office must assume adversarial pressure and build layered controls (not voice alone). ⁵⁶

The overarching design message is simple but nontrivial: **voice AI makes the office more valuable as a place to talk—but only if the office is designed so that talking is productive, private, and trusted.**

¹ ¹⁰ ¹¹ ²³ 2024 Asia Pacific Office Occupier Survey | CBRE

<https://www.cbre.com/insights/reports/2024-asia-pacific-office-occupier-sentiment-survey>

² ¹² Intelligent recap for Teams calls, meetings, and events - Microsoft ...

https://learn.microsoft.com/en-us/microsoftteams/intelligent-recap-calls-meetings?utm_source=chatgpt.com

³ ²⁷ ⁴¹ ⁴⁸ ⁵¹ Employee monitoring – is it right for your business? | ICO

https://ico.org.uk/for-organisations/advice-for-small-organisations/news-blogs-and-events/blogs/employee-monitoring-is-it-right-for-your-business/?utm_source=chatgpt.com

⁴ ⁸ ²² Workspace satisfaction: The privacy-communication trade-off in open ...

https://www.sciencedirect.com/science/article/pii/S0272494413000340?utm_source=chatgpt.com

⁵ ⁴⁴ AI Risk Management Framework | NIST

https://www.nist.gov/itl/ai-risk-management-framework?utm_source=chatgpt.com

⁶ The Influence of Communication Technologies on Organizational Structure ...

https://academic.oup.com/book/52172/chapter/421075009?utm_source=chatgpt.com

⁷ The influence of communication technologies on organizational structure ...

https://dspace.mit.edu/bitstream/handle/1721.1/48153/influenceofcommu00haup.pdf?sequence=1&utm_source=chatgpt.com

⁹ Noise disturbance and lack of privacy: Modeling acoustic ...

https://pubs.aip.org/asa/jasa/article/157/5/3378/3345821/Noise-disturbance-and-lack-of-privacy-Modeling?utm_source=chatgpt.com

¹³ ⁵⁰ Using Meeting Summary with AI Companion - Zoom

https://support.zoom.com/hc/en/article?id=zm_kb&sysparm_article=KB0058013&utm_source=chatgpt.com

¹⁴ Abstractive Meeting Summarization: A Survey - MIT Press

https://direct.mit.edu/tac/article/doi/10.1162/tac_a_00578/116993/Abstractive-Meeting-Summarization-A-Survey?utm_source=chatgpt.com

¹⁵ ¹⁶ Cisco Webex Assistant for Webex Meetings At-a-Glance

https://www.cisco.com/c/en/us/products/collateral/conferencing/webex-meetings/at-a-glance-c45-744053.html?utm_source=chatgpt.com

¹⁷ ²⁸ Multilingual speech recognition in Microsoft Teams

https://support.microsoft.com/en-us/office/multilingual-speech-recognition-in-microsoft-teams-650cb6d2-8a33-40e7-840d-36bb90216aa4?utm_source=chatgpt.com

¹⁸ Use translated captions in Google Meet

https://support.google.com/meet/answer/10964115?co=GENIE.Platform%3DDesktop&hl=en&utm_source=chatgpt.com

¹⁹ Real-time speech-to-speech translation - Google Research

https://research.google/blog/real-time-speech-to-speech-translation/?utm_source=chatgpt.com

20 40 Speaker and Language Recognition | NIST

https://www.nist.gov/programs-projects/speaker-and-language-recognition?utm_source=chatgpt.com

21 [2408.08739] ASVspoof 5: Crowdsourced Speech Data, Deepfakes, and ...

https://arxiv.org/abs/2408.08739?utm_source=chatgpt.com

24 31 32 ISO 3382-3:2022 - Acoustics — Measurement of room acoustic parameters ...

https://www.iso.org/standard/77437.html?utm_source=chatgpt.com

25 29 33 Meeting room guidance for Teams - Microsoft Teams

https://learn.microsoft.com/en-us/microsoftteams/rooms/room-planning-guidance?utm_source=chatgpt.com

26 30 Getting started with Smart Gallery - Zoom

https://support.zoom.com/hc/en/article?id=zm_kb&sysparm_article=KB0066442&utm_source=chatgpt.com

34 Install, Setup, and Deploy Google Meet Hardware - Starter

https://meetingdevices.withgoogle.com/getstarted/?utm_source=chatgpt.com

35 Microsoft Teams Rooms - Microsoft Teams | Microsoft Learn

https://learn.microsoft.com/en-us/microsoftteams/rooms/?utm_source=chatgpt.com

36 BACnet Committee – Website of the BACnet Committee (ASHRAE SSPC 135)

https://bacnet.org/?utm_source=chatgpt.com

37 52 Regulation - 2016/679 - EN - gdpr - EUR-Lex

https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng?utm_source=chatgpt.com

38 39 eur-lex.europa.eu

https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=OJ%3AL_202401689

42 Employment practices: monitoring at work impact assessment

https://ico.org.uk/media2/migrated/4026921/monitoring-at-work-impact-assessment-202310.pdf?utm_source=chatgpt.com

43 45 49 Manage voice recognition for Intelligent Speaker - learn.microsoft.com

https://learn.microsoft.com/en-us/microsoftteams/rooms/voice-recognition?utm_source=chatgpt.com

46 Remotely configure front row layout on Teams Rooms - Microsoft Teams

https://learn.microsoft.com/en-us/microsoftteams/rooms/manage-front-row?utm_source=chatgpt.com

47 Enable Webex Assistant for devices

https://help.webex.com/en-us/article/hzd1aj/Enable-Webex-Assistant-for-Devices?utm_source=chatgpt.com

53 Google Workspace Updates: Speech translation in Google Meet now ...

https://workspaceupdates.googleblog.com/2026/02/speech-translation-meet-ga.html?utm_source=chatgpt.com

54 Dynamic agenda-aware real-time meeting summarization with large ...

https://link.springer.com/article/10.1007/s44443-025-00304-y?utm_source=chatgpt.com

55 ASHRAE 135-2024 | ASHRAE Store

https://store.accuristech.com/ashrae/standards/ashrae-135-2024?product_id=2926783&utm_source=chatgpt.com

56 ISO/IEC 30107-1:2023 - Information technology — Biometric presentation ...

https://www.iso.org/standard/83828.html?utm_source=chatgpt.com